However to give a brief idea, we'd say, it's something like "semi-virtualized" hardware case for the guest OS. In this type, a "hypervisor" is present which can make a "modified" guest OS work on the hardware. It helps improve speed to a substantial degree by allowing few things to be done directly on the hardware rather than implementing full sandbox. For example, if you have a x86 processor on the machine and you're also running a guest OS which can run on the same processor, most hardware can be allowed to be accessed directly (with some restrictions, of course), such as there is no need to emulate the processor as the instructions executed by the guest are okay for the processor to be executed directly. So the hypervisor sees that requests for doing calculations need not be sandboxed or emulated and hence can allow direct access of the processor. This saves the time it takes to emulate the actions and the processing remains fast. Remember however that it does need two mandatory things. Firstly, the guest OS kernel must implement a set of functions which make use of the hypervisor's assistance and that means the guest OS should be modified for the specific case. Secondly, virtualization assistant technologies (e.g. VT-x or AMD-V) should be present in the host machine processor. The "Xen hypervisor" is the best example here.

Advantages of virtualization

We've already discussed this at length, but to summarize, virtualization brings to you various goodies such as: ease of use, scalability for large deployments, failover and fault-tolerance on a system-wide level. It helps ease maintenance and monitoring of the machines and saves cost by hiding the implementation details from the virtual machines. For a general user, it brings an opportunity to experience new and different operating systems, allowing him/her to experiment with the different OSes.

Security

Virtualization mixed with security is a tricky matter as one has to take care of the security of two operating systems, one inside the other. For a general user, it's just okay to secure the host machine. The security scene for Grid, Cluster or Cloud setups with multiple servers is a different scenario and complicated enough. We won't go into that in this Fast Track. However, for a desktop user, it's essential to remember one thing: if you know that either the guest or the host is infected with a virus, make sure that you disconnect the virtual networking interface of the guest OS using the virtualization